

Human Interface Society Review Report

Paper No: S-28-4-002

Date in: 2026/4/26

Reviewer: Reviewer1

1. Individual evaluation for standard items, please grade the paper in 5 levels in terms of three standard items.

1.1. Originality/Novelty

(1) Please mark viewpoint(s) for evaluating the novelty of this paper in the following list.

System

(2) Mark the most appropriate score (and description) in the following list.

2-a: This paper includes few unclear descriptions; however it seems a certain novelty to be published.

(3) Comments: Please justify your evaluation about novelty below.

The author claims that the novelty of this study lies in three contributions: a privacy-preserving browser-native edge architecture, a system design that integrates task-aligned on-device sensing with local data persistence, and an exploratory in-the-wild deployment across laboratory and home settings. However, the claimed novelty of the work should be stated more carefully in relation to existing browser-based and web-deployable physiological sensing platforms. Several prior systems and studies have already explored remote heart-rate detection, browser-based experimental platforms, and online physiological measurement using webcam-based rPPG.

1.2. Availability/usability

(1) Please select and mark one viewpoint for evaluating the availability of this paper in the following list.

a: The paper gives solutions for academic, technical and social needs.

(2) Mark the most appropriate score (and description) in the following list.

2-b: This paper does not show a level of availability/usability, which is considered to be a publishable level. The paper, however, can be published if it has a well-publishable level of novelty.

(3) Comments: Please justify your evaluation about availability/usability below.

The system that the author aims to develop appears to have potential practical value. However, the evaluation method, the number of participant, and the participant characteristics raise substantial concerns. As a result, the evaluation study itself provides only limited useful evidence regarding the effectiveness, or generalizability of the proposed system.

1.3. Reliability

(1) Evaluate the following items based upon “Review standard for reliability” (please mark the appropriate number from 1 to 5)

a. No apparent mistakes and confusion in the manuscript

2

b. Flow of the argument appears straightforward.

2

c. The paper is appropriately justified by reference with relevant study

1

d. The results and outcome of the study are described clearly

2

e. No exaggerated or excessive expressions are found.

1

f. No typos or nonsensical phrase are found.

4

g. The format of the manuscript is suitable for an academic article.

3

h. Experimental conditions and methods are appropriately described.

1

Review standard for reliability

5: Absolutely no problems are present in this evaluation item.

4: It is sufficient level, although it is not perfect.

3: There are minor problems; however, they do not influence the context of the essential part of the paper.

2: There are some problems affecting the context of the essential part of the paper; however they seem to be revised easily.

1: There are some damaging problems affecting the context of the essential part of the paper and they seem to be difficult to correct by the revision.

(2) Comments to the author for its reliability

The reliability of the manuscript is limited by several serious issues. The study evaluates the proposed system with only one participant, and this participant is also the author. Although the manuscript describes the study as an exploratory single-case deployment, this design makes it difficult to assess the reliability or generalizability of the findings. In addition, important details of the experimental conditions are not sufficiently specified. As a result, it is unclear whether the observed differences between the laboratory and home settings reflect the performance of the proposed system, environmental artifacts, or participant-specific effects. Given these limitations, the conclusions are stated too strongly. The manuscript should more clearly separate what can be supported by the current evaluation.

2. Overall evaluation

- Conditional acceptance

3. Comments to the editor

4. Revisions required for acceptance or reasons for rejection

1. The title of the manuscript should be reconsidered.

The manuscript emphasizes that privacy concerns are mitigated by performing data processing locally rather than in the cloud, and therefore uses the phrase “A Privacy-Preserving Browser-Native Framework” in the title. However, the proposed system does not directly implement privacy-preserving techniques, such as those used in prior work on privacy-preserving remote heart-rate estimation from facial videos [1]. Rather, the privacy benefit appears to be an architectural privacy advantage resulting from avoiding raw video transmission and performing processing on the client device, rather than a dedicated privacy-preserving algorithm or method. To avoid giving readers the impression that the paper proposes a dedicated privacy-preserving method, I recommend revising the title to a more modest expression, such as “A Browser-Native Framework for In-the-Wild Physiological Sensing and Task Measurement.”

[1] D. Gupta and A. Etemad, "Privacy-Preserving Remote Heart Rate Estimation from Facial Videos," 2023 IEEE International Conference on Systems, Man, and Cybernetics (SMC), pp. 706-712, 2023. <https://ieeexplore.ieee.org/document/10394350>

2. The necessity of real-time processing is not sufficiently explained.

The manuscript should explain more clearly why real-time video processing is necessary for the proposed system. The current implementation processes webcam images locally and in real time using TensorFlow.js/WebGL, and the manuscript argues that this avoids the need to transmit video data to the cloud. However, given the use cases described in the manuscript

and the evaluation experiment conducted in this study, it is not clear why the video stream must be processed in real time. If accurate timestamps are embedded appropriately, it may also be possible to record the video locally and process it offline after the experiment.

Therefore, the manuscript should clarify the specific role and necessity of real-time processing at an early stage of the paper. For example, real-time processing would be better justified if the system uses physiological or tracking results to adapt the task during the experiment, avoids saving privacy-sensitive raw video by storing only extracted features, or enables online quality control such as detecting tracking failures and adjusting thresholds during data collection. Without such clarification, the real-time processing component appears to be an implementation choice rather than a necessary requirement of the proposed framework.

3. The Research Questions are not defined clearly enough.

The Research Questions should be defined more clearly. In particular, RQ1 asks whether “a browser-native, privacy-conscious sensing framework can be operationally deployed,” but the manuscript does not specify the criteria by which successful deployment is judged. As a result, it is unclear what level of system stability, tracking performance, timing accuracy, or signal quality would be sufficient to conclude that the framework is operationally deployable. The author should clarify the intended deployment scenario and define concrete success criteria that are appropriate for that scenario.

4. It is unclear what the “between-environment differences” in RQ2 are intended to refer to. RQ2 asks what kinds of between-environment differences are observable in the physiological signals collected through the proposed deployment. However, the current design does not allow the reader to determine what these between-environment differences actually mean. Differences between the Lab and Home conditions may reflect changes in the participant’s psychological or physiological state, but they may also reflect differences in the measurement pipeline caused by illumination, luminance fluctuation, screen light, or reduced tracking quality.

This is not merely a limitation of the study, but a central interpretive problem for RQ2 itself. If RQ2 is intended to ask whether physiological signals differ between environments, the present data can only show that estimated signals differed. They cannot show whether those differences represent genuine physiological responses to the environment, measurement

artifacts caused by environmental noise, or a mixture of both. For example, the Home environment differed from the Lab environment in illuminance, luminance stability, tracking confidence, and motion-related noise. Under these conditions, it is difficult to interpret differences in estimated heart rate, or head motion as psychological or physiological effects. Therefore, the manuscript should clearly define the intended meaning of “between-environment differences.” If the authors intend these differences to refer to measurement-quality changes or environmental artifacts, the Results and Discussion should be rewritten accordingly. If the authors intend to interpret them as environment-induced changes in physiological or cognitive state, additional validation is necessary, such as reference physiological measurements, stronger control of lighting/camera/posture conditions, or an experimental design that separates psychological manipulation from environmental measurement noise. Without such validation, the manuscript should avoid interpreting the observed Lab–Home differences as psychological, cognitive, or physiological effects.

5. The novelty and positioning of the study are insufficiently justified

The manuscript should more clearly justify the novelty and positioning of the proposed system. The paper claims three main contributions; however, with respect to the “Privacy-Preserving Edge Architecture,” there appear to be closely related prior studies and systems. For example, prior work has proposed a web application for experimenting with and validating remote measurement of vital signs [2], and commercial or publicly available browser-based rPPG systems such as LabVanced [3] and FacePhys-Demo [4] also provide remote heart-rate estimation from webcam video. In addition, RhythmEdge explicitly addresses contactless heart-rate estimation on edge devices [5]. These examples suggest that browser-based, web-deployable, or edge-based rPPG systems have already been explored to some extent.

Therefore, the Related Work section should be revised to include these relevant studies and systems, and the manuscript should clearly explain how the proposed framework differs from them. If the authors cannot identify a clear technical or conceptual distinction, the claim of novelty regarding the privacy-preserving edge architecture should be withdrawn or substantially weakened.

[2] A. H. Ayesha, D. Qiao, and F. Zulkernine, “A web application for experimenting and validating remote measurement of vital signs”, *Information Integration and Web Intelligence (iiWAS 2022)*, Lecture Notes in Computer Science, vol. 13635, 2022.

https://link.springer.com/chapter/10.1007/978-3-031-21047-1_21

[3] LabVanced, <https://www.labvanced.com/content/technology/en/remote-heart-rate-detection-rPPG/>

[4] FacePhys-Demo <https://github.com/KegangWangCCNU/FacePhys-Demo>

[5] Z. Hasan, E. Dey, S. R. Ramamurthy, N. Roy and A. Misra, "Demo: RhythmEdge: Enabling Contactless Heart Rate Estimation on the Edge," 2022 IEEE International Conference on Smart Computing (SMARTCOMP), pp. 177-179, 2022.

<https://ieeexplore.ieee.org/document/9821075>

Regarding the claimed contribution of the “Exploratory In-the-Wild Deployment,” the manuscript argues that the deployment demonstrates that heterogeneous real-world environments affect the collected physiological signals. However, the fact that environmental conditions influence rPPG measurements in the wild is already well known and has been discussed in prior work, including studies on rPPG using online webcams in uncontrolled settings[6]. Therefore, simply showing that environmental differences affect the measurement results does not appear to constitute a sufficiently strong contribution by itself. If the authors wish to retain this point as one of the main contributions, the manuscript should go beyond demonstrating the mere existence of environmental effects. It should instead clarify how robustly the proposed framework operates under such heterogeneity, which measurements remain reliable across environments, and which measurements fail or become unreliable under specific environmental conditions.

[6] Di Lernia, D., Finotti, G., Tsakiris, M. et al. “Remote photoplethysmography (rPPG) in the wild: Remote heart rate imaging via online webcams.”, *Behav. Res.*, Vol.56, pp.6904–6914, 2024.

On the other hand, the contribution related to “System Design for Task-Aligned On-Device Sensing and Local Persistence” may still constitute a meaningful novelty. To the best of my knowledge, there are few, if any, prior papers that explicitly present a system integrating psychophysical task execution, rPPG/FaceMesh-based physiological feature extraction, and timestamped local log persistence within the same browser runtime. Therefore, the authors may be able to claim novelty in this integrated system design, rather than in local rPPG processing or privacy-preserving edge processing alone.

However, this point should be stated more explicitly and carefully in the manuscript. The authors should clarify that the main contribution is the integration of task-aligned sensing, on-device feature extraction, and local-first logging in a unified browser-native runtime, and should distinguish this contribution from prior browser-based or edge-based rPPG systems.

6. Unclear claim of robustness against rigid head motion

In Section 2.1, the manuscript identifies performance degradation of rPPG under real-world conditions, such as illumination changes and motion, as a key challenge. However, immediately after this, the authors state that the proposed framework prioritizes computational efficiency and robustness against rigid head motion. The relationship between the stated problem and this design choice is unclear.

In particular, the manuscript does not explain what algorithmic mechanism or implementation strategy provides robustness against rigid head motion. It is also unclear whether the proposed framework is intended to mitigate rPPG degradation in real-world environments, to merely observe and record such degradation, or to accept reduced measurement accuracy as a trade-off for privacy preservation and real-time browser-native operation.

The authors should clarify the intended role of the proposed framework with respect to rPPG robustness in real-world settings. If robustness is claimed, the manuscript should explain how it is achieved and provide appropriate evidence. If robustness is not claimed, the relevant descriptions should be revised so that the framework is presented as a deployment and logging architecture rather than as a method for improving rPPG signal quality.

7. Insufficient description of the concrete usage scenario.

The manuscript should clarify the concrete usage scenario of the proposed system earlier in the paper. At present, it is difficult to understand whether Cameralab is intended for remote psychophysical experiments conducted at home, for continuous physiological sensing during daily activities, or for some other deployment scenario.

8. Definition of “in-the-wild” and “naturalistic physiological sensing” are not clear.

The manuscript repeatedly uses expressions such as “in-the-wild” and “naturalistic physiological sensing.” However, the actual deployment described in the paper consists of a participant sitting in front of a laptop, viewing Gabor stimuli on the screen, and responding

by key press in home and laboratory environments. This is a highly task-constrained home/lab experiment and is substantially different from unconstrained continuous physiological sensing during walking or everyday activities.

The authors should clearly specify the target use case of Cameralab. If the intended scenario is a remote psychophysical experiment with physiological sensing conducted in home environments, terms such as “in-the-wild” and “naturalistic” should be used more cautiously and with appropriate qualification. The manuscript should avoid giving the impression that the framework has been evaluated for unconstrained daily-life physiological sensing.

9. The description of the system hardware is insufficient for reproducibility.

The manuscript does not provide enough information about the PC and camera used in the experiment. Since the study evaluates the feasibility of real-time browser-based FaceMesh inference and rPPG extraction, the authors should report the relevant hardware and software specifications in detail. These include, at minimum, the PC model, CPU/GPU, RAM, operating system, browser name and version, webcam type, camera resolution, nominal frame rate, and camera settings such as auto-exposure and auto-white-balance.

10. The description of the experimental environments is insufficient.

The environmental conditions are not described in enough detail to interpret the reported differences between the Lab and Home settings. Although the mean illuminance values are reported, the manuscript does not specify important physical factors such as the presence of windows, contribution of natural light, time of day, weather, or curtain conditions. In particular, the text mentions clouds, screen flicker, and power cycles as possible sources of luminance fluctuation, but it is unclear whether these factors were actually present in the Home sessions.

The measurement protocol for illuminance and luminance fluctuation should also be clarified. For example, the authors should explain whether the reported lux values represent within-session variation, between-session variation, or measurements over a specific time window. Similarly, the aggregation windows for Mean Luminance and Exposure Fluctuation should be specified.

11. The validity of the cognitive framing manipulation is questionable.

Because the participant was the author, the participant necessarily knew that the Negative

Evaluation and Positive Evaluation instructions were experimentally constructed and did not involve any real penalty, reward, or external evaluation. Therefore, it is difficult to assume that these instructions induced evaluation pressure or reward motivation.

Accordingly, the manuscript should not interpret the observed differences as evidence of cognitive or motivational effects caused by the framing manipulation. Rather, the authors should reposition these conditions as an exploratory demonstration that the system can synchronize physiological sensing and behavioral task measurement across different task contexts or instructional conditions.

12. The central claim in the Discussion should be reconsidered.

The manuscript presents as a main finding that laboratory-only validation may overestimate signal stability and condition separability in home use. However, this point appears to largely restate a known limitation of video-based physiological sensing, namely that signal quality is affected by lighting, posture, motion, and camera placement. Indeed, the manuscript itself already identifies these factors as part of the motivation in the Introduction.

Therefore, the authors should clarify what was newly demonstrated by the present study beyond this known problem. For example, the Discussion could focus more specifically on what design requirements or safeguards should be incorporated into future versions of Cameralea.

13. Several claims in the manuscript are not sufficiently supported by the current evidence and should be substantially weakened.

First, in the final part of the Limitations section, the manuscript states that the present study “establishes” a concrete browser-native design pattern for privacy-conscious, task-aligned sensing. This wording is too strong. The study is based on an exploratory N=1 deployment, the participant was the author, no comparison with existing runtimes or APIs was conducted, and reproducibility across multiple devices, participants, or environments has not been demonstrated. Therefore, the manuscript should avoid claiming that it has “established” such a design pattern. A more appropriate interpretation would be that the study presents a browser-native implementation example of privacy-conscious, task-aligned sensing under limited Lab/Home conditions.

Second, the claim that the Home deployment was feasible because “the tracking confidence remained above 96% at home” is not sufficiently justified. This value is likely to be strongly

affected by the exclusion criterion of $\text{Conf} < 0.8$, but the rationale for this threshold is not explained. The manuscript should clarify why 0.8 was chosen, how MediaPipe tracking confidence relates to rPPG signal quality or landmark accuracy, and what level of tracking confidence is sufficient to support a claim of feasibility. Without such justification, the 96% tracking-confidence result alone cannot be used as strong evidence of deployment feasibility. Third, the scope of the term “naturalistic deployment” should be carefully limited. Although the study was conducted in a Home environment, the participant was sitting in front of a laptop, looking at the screen, and performing an experimental task. This situation is substantially different from ordinary daily-life activities such as cooking, walking, conversation, night-time use, or situations with large illumination changes. Moreover, the participant was the author, who presumably understood the purpose of the system and may have been more likely than ordinary users to keep the face within the camera view, maintain posture, and cooperate with the task procedure. These factors may lead to an overestimation of practical deployability.

Accordingly, the manuscript should revise claims about feasibility and naturalistic deployment more conservatively. Based on the current data, the study can only support the limited conclusion that, for this specific author-participant, in this particular home setting, time period, posture, laptop/camera configuration, lighting condition, and task-execution context, FaceMesh tracking did not substantially fail. Stronger claims about general feasibility, robustness, or naturalistic deployment are not warranted by the present evidence. Abstract and conclusions should also be revised accordingly.

5. Other comments

The readability of some visual elements should be improved. In particular, the text in Table 1, Figure 2, and Figure 4 is too small and difficult to read. The author should enlarge the font size and revise the layout so that the contents can be understood more easily. In addition, Figure 2 is difficult to interpret because the manuscript does not provide a sufficient explanation of how the figure should be read. The author should add a clearer description in the main text or the figure caption.

The caption of Figure 3 appears to be inconsistent with the figure itself. The caption states

that the Home environment exhibits “higher exposure fluctuation and lower tracking confidence compared to the Laboratory,” but tracking confidence does not seem to be shown in Figure 3. The author should revise the caption to describe only the variables actually shown in the figure.